# TELCO CUSTOMER CHURN ANALYSIS REPORT

## SITUATION

The telecom company was experiencing a high customer churn rate. They wanted to understand **why customers are leaving** and which customer segments are at the highest risk.
The dataset contained **7043 customer records** with details like tenure, internet services, contract type, billing method, monthly charges, etc.

## TASK

My goal was to:

1. **Clean the raw dataset** using SQL (fix data types, remove blanks, handle missing values).
2. Perform **Exploratory Data Analysis (EDA)** in Python to identify churn drivers.
3. Build a **Power BI dashboard** to present findings visually and help stakeholders make decisions.

The goal was to give clear, actionable insights like:

- *Which type of customers churn the most?*
- *Which services or billing methods increase churn probability?*

## ACTION

I executed the project in 3 phases:

**1. Data Cleaning (SQL)**

- Checked duplicates, blanks, incorrect data types, and inconsistent values.
- Fixed invalid values in Total Charges and converted them to numeric format.
- Exported a clean dataset into Telco_Churn_Clean.csv.

**2. Exploratory Data Analysis (Python + Pandas + Seaborn/Matplotlib)**

- Aggregated churn counts by:
  - Contract type
  - Internet service
  - Payment method
  - Tenure categories
- Created tenure bins to analyze churn rate across customer lifecycle.
- Added new features (example: total services used).

Key insights discovered:

- *Month-to-month customers churn the most.*
- *Electronic check users have the highest churn.*
- *Churn decreases as customer tenure increases.*
- *Customers using fiber optic and without online security churn heavily.*

**3. Dashboard (Power BI)**

Built an interactive dashboard with:

- **5 KPIs** (Total Customers, Churned Customers, Churn Rate, Avg Monthly Charges, Avg Tenure)
- **Slicers** (Contract, Internet Service, Payment Method)
- **7 visualizations** including:
  - Churn by Contract Type
  - Churn by Internet Service
  - Churn by Payment Method
  - Total Monthly Charges by Churn
  - Churn by Tenure category (line chart)
  - Services used vs churn (matrix)

Dashboard demonstrated:

- Who churns
- Why they churn
- What business actions can reduce churn

## RESULT

- The analysis revealed very clear churn-driving patterns.
- Customers on ***month-to-month contracts*** **showed the highest churn rate**, while customers with longer-term contracts (one or two years) rarely left the company.
- A strong correlation was also observed between churn and payment methods — customers paying through **Electronic Check churned significantly more** than those using auto-debit or credit cards.
- **Internet service** type also played a major role, with **Fiber-optic users churning the most**, especially when they lacked additional support services such as Online Security and Tech Support.
- **Tenure** was another major differentiator: **churn rates were high during the initial few months but dropped steadily for customers who stayed beyond one year.**
- These insights indicate that early retention strategies, plan bundling, and encouraging customers to shift to auto-payment or long-term contracts could significantly reduce churn.